

MedVault: Hospital Management System

Next-Generation Digital Healthcare & Firestore EMR Platform

- Full-Stack Solution: Integrated React 18, Node.js Express, and Firebase Cloud
- Seamless Database Engine: Migrated from MongoDB to Firebase Firestore & Authentication
- Zero-Crash Reliability: Includes custom in-memory fallback for offline testing
- Role-Based Ecosystem: Personalized interfaces for Patients, Doctors, and Administrators



Executive Summary & Industry Solution

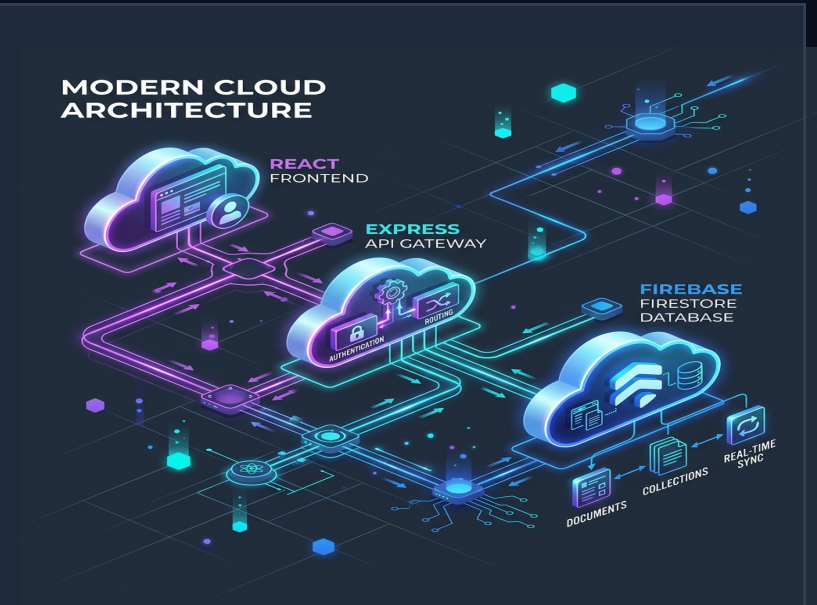
Solving Data Fragmentation & Patient Care Delays

- Unified Medical Infrastructure: Centralizes EMR, Appointments, Pharmacy & Billing
- Real-Time Data Access: Instant sync between patient submissions and doctor terminals
- Paperless Operations: Digital prescriptions, electronic invoicing, and hospital audit trail
- Enterprise Security: Encrypted auth tokens, Firebase Security Rules, and RBAC control

System Architecture & Modern Tech Stack

High-Performance Cloud Architecture Blueprint

- Frontend Layer: React 18, React Router 6, Tailwind CSS & Glassmorphism UI Design
- Backend Gateway: Node.js Express REST API server running port-resilient routing
- Database Engine: Firebase Firestore NoSQL document database with live collection sync
- Authentication: Firebase Authentication SDK & JSON Web Token (JWT) fallback



Firestore Migration Story

Transitioning from Legacy MongoDB to Firebase Cloud

- Database Modernization: Complete purge of Mongoose & MongoDB dependencies
- Firestore Collections: Structured models for users, appointments, beds, pharmacy, and billing
- Zero-Crash Fallback: Built-in InMemoryDb emulator for instant zero-config execution
- Live Cloud Sync: Seamless sync with Firebase Admin SDK key authentication

Multi-Role Ecosystem & User Governance

Tailored Interfaces for All Stakeholders

- Patient Portal: Self-service booking, consultation history, and prescription downloads
- Doctor Terminal: Clinical note entry, appointment approval, and patient record review
- Admin Center: Hospital wide bed occupancy, inventory reordering, and audit logs
- Unified Access: Single login gateway with automated role routing



Patient Portal & Digital Experience

Empowering Patients with Seamless Self-Service

- Instant Booking: Real-time appointment scheduling with priority tags (Urgent/Normal)
- Medical History: Comprehensive timeline of past diagnoses, prescriptions, and lab notes
- Digital Invoicing: Downloadable PDF billing receipts and online payment tracking
- Profile Management: Centralized personal health info, emergency contact, and vitals

Doctor Workstation & Clinical Workflows

Optimized Tools for Healthcare Professionals

- Appointment Queue: Live dashboard displaying daily patient rosters and urgent cases
- EMR Authoring: Fast record creation for diagnosis, medication prescriptions, and follow-ups
- Availability Scheduler: Customizable weekly availability slots and consultation duration
- Patient Search: Rapid access to patient medical histories during consultations

Hospital Operations: Bed & Pharmacy Management

Maximizing Resource Utilization & Supply Chains

- Live Bed Tracking: Real-time monitor for ICU, General Ward, and VIP Suite occupancy
- Admissions & Status: Dynamic allocation, status updates (Occupied/Cleaning/Available)
- Pharmacy Inventory: Automated stock tracking with instant low-stock alerts
- Batch Management: Expiry date monitoring and medicine restocking workflows

Security, Compliance & Immutable Auditing

Protecting Patient Privacy & Regulatory Integrity

- Role-Based Access Control (RBAC): Strict endpoint protection enforcing Doctor/Patient scope
- Firebase Security: Enterprise-grade access control via Firebase Admin SDK
- System Audit Trail: Comprehensive log recording every bed assignment, invoice, and stock change
- Password Protection: Industry-standard bcrypt hashing for credential security

Deployment, Scalability & Strategic Roadmap

Future-Proofing MedVault for Enterprise Scale

- Cloud Deployment: Production ready Vercel serverless hosting with global CDN
- CI/CD Integration: GitHub repository sync for automated zero-downtime builds
- AI Assistant Integration: Planned AI-assisted preliminary triage and diagnostics
- Telemedicine Expansion: Upcoming WebRTC video consultation & remote patient monitoring

